Solar supercapacitor could power future wearable sensors

Currently, wearable systems generally rely on relatively heavy, inflexible batteries, which can be uncomfortable for long-term users. Researchers from University of Glasgow's Bendable Electronics and Sensing Technologies (BEST) group have developed a graphene supercapacitor, which could be used in the next generation of wearable health sensors.

Led by Prof. Ravinder Dahiya, the team built on its previous success in developing flexible sensors by developing a supercapacitor that could power health sensors capable of conforming to wearer's bodies, offering more comfort and a more consistent contact with skin to better collect health data.

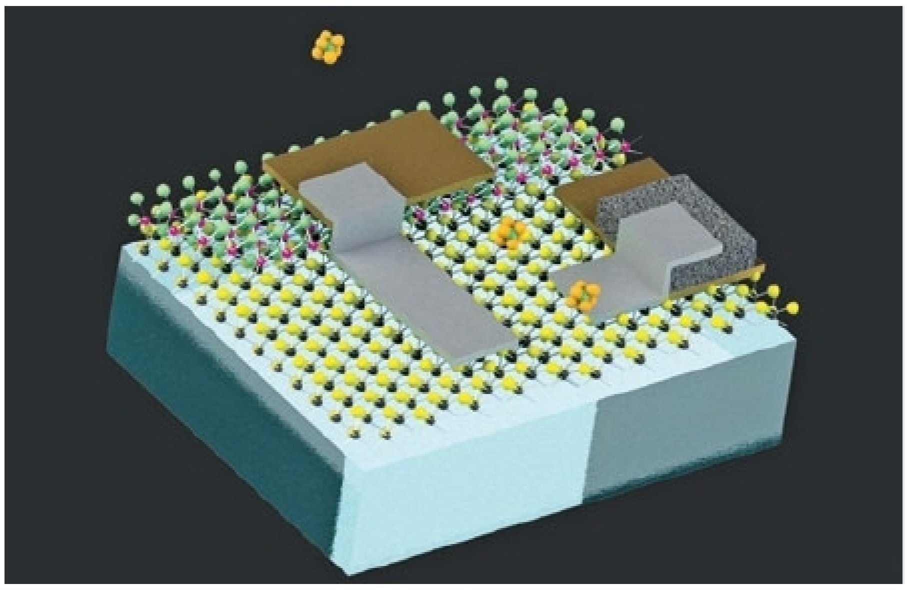
The new supercapacitor uses layers of flexible, three-dimensional porous foam formed from graphene and silver to produce a device capable of storing and releasing around three times more power than any similar flexible supercapacitor. The team demonstrated the durability of the supercapacitor, showing that it provided power consistently across 25,000 charging and discharging cycles.

The researchers also found a way to charge the system by integrating it with flexible solar-powered skin already developed by the BEST group, effectively creating an entirely self-charging system, as well as a pH sensor that uses wearers' sweat to monitor their health.

Prof. Dahiya says, "This research could take wearable systems for health monitoring to remote parts of the world where solar power is often the most reliable source of energy. It could also increase the efficiency of hybrid electric vehicles. We are looking at further integrating the technology into flexible synthetic skin, which we are developing for use in advanced prosthetics."

Cell-sized robots that can sense their environment

Researchers at Massachusetts Institute of Technology (MIT) have created what may be the smallest robots yet that can sense their environment, store data and even carry out computational tasks. These devices are about the size of a human egg cell and consist of tiny electronic circuits made of two-dimensional materials, piggybacking on minuscule particles called colloids.



Made of electronic circuits coupled to minute particles, the devices could flow through intestines or pipelines to detect problems (Credit: Massachusetts Institute of Technology)

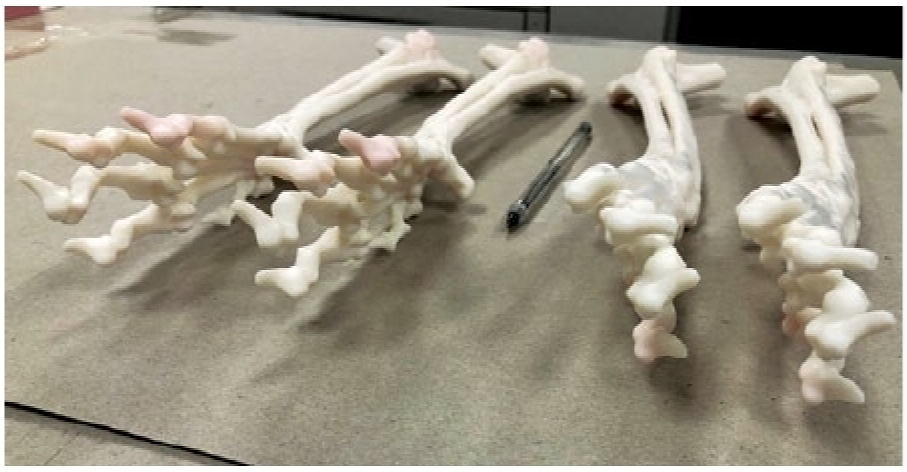
Colloids—insoluble particles or molecules measuring anywhere from a billionth to a millionth of a metre across—are so tiny that these can stay suspended indefinitely in a liquid or even in air. By coupling these tiny objects to complex circuitry, researchers hope to lay the groundwork for devices that could be dispersed to carry out diagnostic journeys through anything from the human digestive system to oil and gas pipelines, or perhaps to waft through air to measure compounds inside a chemical processor or refinery.

Michael Strano, Carbon C. Dubbs Professor of Chemical Engineering at MIT and senior author of the study explains, "We wanted to figure out methods to graft complete, intact electronic circuits onto colloidal particles."

These tiny robots are self-powered and require no external power source or even internal batteries. A simple photodiode provides the trickle of electricity that the robots' circuits need to power their computation and memory circuits. This is enough to let them sense information about their environment, store that data in their memory and then later have the data read out after accomplishing their mission.

3D printing of body parts to commercialise medical training

3D-printed human body parts with life-like bone, skin and muscle densities are being developed in South Australia as teaching aids for surgical training. Medical devices can also be designed with built-in pathogens, such as tumours, bone fractures or defective hearts, to allow surgeons and students to practise specific procedures.



Replica dog leg bones to help veterinary surgeons plan an operation to correct bowed legs in a dog (Credit: The Lead South Australia)

According to Mark Roe, founder - Fusetec 3D, the main benefit with this machine is that, human tissue density can be replicated. Students can operate on it, sew it and drill bones, so it simulates a human cadaver. Density from bone-like to skin-like to muscle-like can all be changed in the same print job.

Roe says that although similar technologies are being used for research and patient-specific modelling at universities and medical centres in North America and Europe, his system is the only purely commercial operation of its kind in the world.